

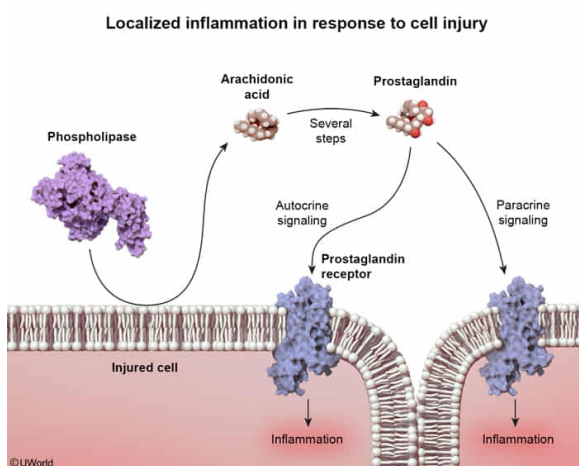
Phospholipases release arachidonic acid from cell membranes as part of the inflammatory response. Given this information, which type of molecule is derived from arachidonic acid and is involved in this response?

- ☐ A. Terpenes [11%]
- ☐ B. Ketone bodies [7%]
- ☐ C. Catecholamines [13%]
- ✔ ☒ D. Prostaglandins [68%]

Correct

67% answered correctly

Explanation:



Localized inflammatory responses occur near the site of an injury or infection. They frequently involve phospholipase enzymes releasing arachidonic acid from the plasma membranes of affected cells. Arachidonic acid is then modified by cyclooxygenase and a series of other enzymes to form various 20-carbon (**eicosanoid**), nonhydrolyzable lipids known as **prostaglandins**. Prostaglandins act as **autocrine** and **paracrine** signals (binding to receptors on or near the cell from which they originated). Because they act only on nearby cells, prostaglandins produce a **localized inflammatory response**. Nonsteroidal anti-inflammatory drugs (NSAIDs) such as aspirin and ibuprofen treat inflammation by inhibiting prostaglandin synthesis.

(Choice A) **Terpenes** are precursors in the synthesis of cholesterol and steroid hormones. They are derived from isoprenes and do not play a role in inflammatory responses.

(Choice B) **Ketone bodies** are fuel sources that are produced under low glucose conditions. They are derived from acetyl-CoA and are not involved in inflammatory responses.

(Choice C) **Catecholamines** such as epinephrine and norepinephrine are hormones derived from the amino acid tyrosine. They can play a role in inflammatory responses but

generally do so as **endocrine signals** (acting on cells that are distant from their origin).

Educational objective:

Prostaglandins are nonhydrolyzable, 20-carbon (eicosanoid) lipids involved in autocrine and paracrine signaling. They are derived from arachidonic acid and often help mediate localized inflammatory responses.

Time spent:0 sec

Subject:
Biochemistry

QID:401913

System:
Carbohydrates, Nucleotides, and Lipids